

Demonstration of Proposed Features

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	1 Mark

Feature Delivery & Demonstration Verification:

This document records all features proposed during the project planning phase and tracks their successful implementation and live demonstration.

S.No	Feature Name	Description & Implementation	Status	Demo'd	Verification Remarks
1	User Registration & Login	Registers account, salts password via bcrypt, issues 7-day HMAC-SHA256 JWT.	Done	Yes	Demonstrated live in Postman with 201/200 OK responses.
2	JWT Protected Profile	auth.js extracts Bearer token, fetches user profile without password.	Done	Yes	Verified token decoding and 401 on missing token.
3	Workout Creation & Logging	Logs activity with category, duration, calories, and timestamp.	Done	Yes	Verified in MongoDB Atlas cluster with new ObjectId.
4	View All Workouts	Retrieves all user workouts sorted chronologically by workoutDate descending.	Done	Yes	Verified user isolation and date sorting.
5	Multi-Criteria Search	Searches workouts via case-insensitive regex query on name, category, or date.	Done	Yes	Verified ?q=running and ?q=jog returning matching arrays.
6	Update & Delete Workouts	Modifies or deletes workout records with strict ownership validation.	Done	Yes	Verified 200 OK on owner and 404/403 on non-owner.
7	AI Workout Recommendation	Google Gemini 1.5 Flash generates tailored workout routine from user inputs.	Done	Yes	Generated complete routine in 1.42s live.
8	AI Fitness Insights	Gemini AI analyzes workout totals to provide actionable progress feedback.	Done	Yes	Generated tailored consistency feedback in 1.35s live.